

MT AIRY, NC, USA

WMO: 723177

Lat: 36.460N

Lon: 80.553W

Elev: 380

StdP: 96.84

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -5.1	(d) -22.3	(e) 0.5	(f) 0.5	(g) -18.8	(h) 0.7	(i) 0.3	(j) 7.9	(k) 4.0	(l) 7.1	(m) 4.9	(n) 1.5	(o) 320	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	33.0	23.3	32.2	23.1	31.0	23.0	25.6	30.5	24.9	29.7	24.2	28.8	1.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.8	28.5	23.3	18.9	27.8	22.7	18.3	27.1	81.1	30.9	77.6	29.7	75.2	28.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.7	5.4	4.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-10.6	35.1	2.6	1.6	-12.5	36.2	-14.0	37.2	-15.5	38.1	-17.4	39.2		
			-12.1	26.3	2.3	0.8	-13.8	27.0	-15.1	27.4	-16.4	27.9	-18.1	28.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	4.0	5.6	9.9	14.9	19.1	23.5	25.3	24.7	21.4	15.4	9.8	6.1		
	DBStd	8.49	5.27	4.84	4.92	4.31	3.82	2.65	2.19	2.45	3.10	4.27	4.62	4.37		
	HDD10.0	609	196	137	64	9	0	0	0	0	0	7	59	137		
	HDD18.3	1978	443	357	262	116	39	2	0	0	10	111	257	381		
	CDD10.0	2444	11	14	62	157	283	404	473	457	342	173	53	15		
	CDD18.3	771	0	1	2	15	63	156	214	199	102	18	1	0		
	CDH23.3	6353	1	1	22	180	504	1291	1873	1650	682	135	14	0		
	CDH26.7	2168	0	0	1	33	127	447	734	614	188	24	1	0		
(14) Wind	WSAvg	1.4	1.6	1.7	2.0	1.9	1.4	1.1	0.9	0.9	1.3	1.3	1.4	1.3		
(15) Precipitation	PrecAvg	1225	95	87	113	97	113	108	125	112	109	90	86	89		
	PrecMax	1777	196	185	250	227	210	254	363	318	287	284	235	210		
	PrecMin	820	20	12	20	16	30	17	38	12	5	0	15	8		
	PrecStd	199	42	43	51	47	46	61	59	70	66	62	42	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.2	26.1	29.9	32.1	33.9	34.9	35.2	32.4	29.2	25.9	21.3		
		MCWB	14.4	15.8	15.0	16.5	21.5	22.9	24.3	23.0	22.2	22.4	16.5	14.9		
	2%	DB	17.5	19.0	23.0	27.4	29.2	32.3	33.0	32.8	30.5	26.6	22.4	17.8		
		MCWB	12.4	13.5	14.1	16.6	20.7	22.8	24.1	23.2	22.3	18.7	14.9	13.6		
	5%	DB	15.3	16.8	21.2	25.6	27.9	31.0	32.2	31.9	28.8	24.7	20.2	15.9		
		MCWB	10.8	10.6	13.2	16.4	20.2	22.5	23.7	23.1	21.8	18.2	14.3	11.9		
	10%	DB	12.5	13.9	18.5	23.0	26.4	29.1	30.9	30.1	27.4	22.7	17.9	13.2		
		MCWB	8.5	8.3	11.9	15.0	19.4	21.8	23.4	23.3	20.6	17.5	12.9	9.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	17.6	17.8	20.9	23.9	25.3	26.6	26.8	24.7	23.6	19.2	18.3		
		MCDB	18.5	20.8	22.9	24.5	28.9	30.4	31.4	32.0	29.2	27.3	22.2	19.8		
	2%	WB	14.0	15.1	16.2	19.1	22.6	24.5	25.7	25.7	24.0	21.8	17.5	15.3		
		MCDB	16.8	17.7	20.1	23.8	27.3	29.6	30.8	30.4	28.5	24.6	19.9	17.0		
	5%	WB	11.6	11.9	15.1	17.9	21.6	23.8	25.0	24.8	23.1	20.0	15.7	12.6		
		MCDB	14.0	15.2	19.3	22.9	26.2	28.8	30.1	29.2	27.0	23.2	19.0	14.5		
	10%	WB	9.1	9.2	13.5	16.8	20.6	23.1	24.3	24.0	22.1	18.3	13.8	10.3		
		MCDB	11.8	12.7	17.5	21.7	25.0	27.7	29.1	28.2	25.4	21.3	17.1	12.6		
(35) Mean Daily Temperature Range	MDBR		9.9	10.3	11.2	12.0	10.9	10.6	9.9	9.6	9.7	10.8	11.3	9.3		
	5% DB	MCDBR	13.5	14.4	14.5	15.2	12.8	12.3	11.6	11.6	12.0	13.2	13.8	12.3		
		MCWBR	8.9	9.3	7.5	7.0	6.1	5.2	4.8	4.6	5.6	6.8	7.7	8.3		
	5% WB	MCDBR	11.0	12.4	12.1	11.8	10.7	10.6	10.2	9.6	9.7	10.0	10.1	9.7		
		MCWBR	9.0	10.0	7.8	6.6	5.8	5.2	4.9	4.7	5.2	6.4	7.2	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.317	0.344	0.390	0.430	0.466	0.504	0.504	0.401	0.359	0.318	0.304		
	taud		2.493	2.450	2.420	2.304	2.210	2.147	2.068	2.043	2.330	2.434	2.505	2.527		
	Ebn at Noon		895	922	923	893	857	823	788	778	851	862	869	872		
	Edh at Noon		81	96	108	128	143	153	164	165	117	96	80	74		
(44) All-Sky Solar Radiation	RadAvg		2.47	3.12	4.18	5.21	5.73	6.18	5.73	5.27	4.43	3.63	2.75	2.10		
	RadStd		0.17	0.38	0.26	0.42	0.47	0.45	0.37	0.28	0.41	0.45	0.23	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+87

Nomenclature: See separate page